Event Density: Open Note for Experiments

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February 2026

This is a living document of empirical predictions, testable consequences, and experimental paths for the Event Density (ED) framework.

This Open Note collects clean, falsifiable, publishable experimental tests of the Event Density ontology. Each section is a short, self-contained experimental proposal designed to confirm or break a specific ED prediction.

This document will expand as new predictions and experimental paths are developed.

Expanded and detailed outline of all experiments are available at:
https://github.com/allen-proxmire/event-density/tree/main/ED%20Testing

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Quantum Mechanical Tests

1. ED Decoherence Rule

Introduction
ED predicts that decoherence is governed not by mass or size, but by the gradient-complexity of the event-density field. Symmetric systems maintain coherent participation; asymmetric systems decohere rapidly.

Prediction
The decoherence rate scales with ED-complexity:

$$\Gamma_{\text{decoh}} = k \int \| \nabla \rho(x) \|^2 \, d^3x$$

- High symmetry → low ED-complexity → long coherence
- High asymmetry → high ED-complexity → rapid decoherence

Experimental Path
Construct a structural-complexity ladder:

- C_{60} (buckyball)
- C_{70}
- rigid asymmetric organic molecule
- floppy biomolecule (peptide)
- virus capsid

For each:

1. Compute C_{ED} from structural data
2. Measure coherence time in interferometry
3. Compare ordering and scaling

A clean deviation from mass-based scaling is a decisive test.

2. Biological Coherence Limit

Introduction
ED predicts a strict upper bound on quantum coherence for biological structures due to their extreme ED-complexity.

Prediction
Above a critical ED-complexity threshold, biological structures cannot remain quantum.

Consequences:
- microtubules decohere essentially instantly
- proteins decohere too fast for quantum-biological effects
- viruses are borderline but should fail to show interference

Experimental Path
- Perform interferometry on peptides, proteins, and virus capsids
- Compute ED-complexity for each
- Look for the predicted threshold behavior

A single positive interference result for a high-complexity biomolecule would falsify this ED prediction.

3. Interference Visibility Scaling

Introduction
Standard QM predicts exponential decay of interference visibility with mass or environmental coupling. ED predicts a different scaling law.

Prediction
Visibility decays as a power-law in ED-complexity:

$$V \sim 1 / 1+\alpha C_{ED}$$

not exponentially in mass.

Experimental Path
- Prepare molecules with controlled structural complexity
- Measure interference visibility
- Fit both models:
    - ED power-law
    - standard exponential

Whichever model fits better is the winner.

4. Hyper-Coherence in Ultra-Symmetric Systems

Introduction
ED predicts that systems with extremely low ED gradients exhibit coherence times longer than standard QM expects.

Prediction
Ultra-symmetric systems should show anomalously long coherence:

- superconducting loops
- Bose–Einstein condensates
- symmetric nanoparticles
- perfectly symmetric optical lattices

Experimental Path

- Measure coherence times in ultra-symmetric engineered systems
- Compare to standard decoherence models
- Look for positive anomalies predicted by ED

A single confirmed anomaly is a strong ED win.

5. Quantum–Classical Threshold

Introduction
Standard QM treats decoherence as a smooth crossover. ED predicts a sharp, thresholded transition based on ED-complexity.

Prediction
There exists a critical ED-complexity C_{crit}:

- below C_{crit}: stable quantum behavior
- above C_{crit}: immediate classicality

Experimental Path

- Incrementally increase molecular complexity
- Track coherence time
- Identify the predicted sharp transition

This is a clean falsification target.

6. Bell Correlation Degradation with Complexity

Introduction
ED predicts that entanglement strength depends on ED-complexity, not just environmental coupling.

Prediction
Bell-type correlations degrade as ED-complexity increases, even with environmental coupling held constant.

Experimental Path

- Send entangled photons through structured media

- Use entangled ions with increasing internal complexity
- Measure correlation strength vs. ED-complexity

A complexity-dependent degradation is a decisive ED signature.

7. Quantum Error Correction Limit

Introduction
ED predicts a fundamental limit to scalable quantum error correction based on the ED-complexity of encoded states.

Prediction
Quantum error correction fails above a complexity threshold because high-complexity ED patterns cannot maintain coherent participation.

Experimental Path
- Perform multi-qubit scaling tests
- Track error-correction fidelity vs. ED-complexity
- Identify the predicted breakdown point

This is a long-term but high-impact test.

Galactic and Cosmological Tests and Observations

8. Dwarf Galaxy Rotation Curves (Completed Test — Passed)

Introduction
ED predicts that galaxy rotation curves follow event-density tension, not dark-matter halo profiles. In particular, ED makes a clean, falsifiable prediction:

> Dynamically Active dwarf galaxies should exhibit larger outer-radius mass discrepancies than Quiet dwarfs, even at fixed baryonic mass.

This prediction arises directly from ED's activity-dependent temporal tension: sustained internal activity generates ED gradients that diffuse into smooth temporal halos, increasing apparent curvature.

Status
This prediction has now been tested and passed.
A full empirical analysis using the public SPARC dataset (46 dwarf galaxies) shows:
Quiet dwarfs:
- $\langle D_{\text{outer}} \rangle \approx 3.94$

Active dwarfs:
- $\langle D_{\text{outer}} \rangle \approx 6.01$

Ratio:
- $D_{\text{Active}} / D_{\text{Quiet}}$t ≈ 1.53

Active dwarfs exhibit a 53% higher outer-radius mass discrepancy than Quiet dwarfs.
A scatter plot of the outermost points shows two clean vertical bands: Quiet dwarfs at D ≈ 3–4, Active dwarfs at D ≈ 5–6.

The separation is structural, reproducible, and visually obvious.

This is exactly the ED prediction:
increased internal activity → increased temporal tension → increased apparent curvature.

Prediction (Confirmed)
ED predicts:

- a specific scaling between baryonic distribution and rotational tension
- a "tension plateau" in dwarfs
- deviations from NFW halo fits in low-mass systems
- a mass-discrepancy curve that follows ED tension
- Active > Quiet in outer-radius discrepancy

Experimental Path (Completed)
The following steps were executed:

- pulled SPARC rotation-curve data
- filtered to 46 dwarf galaxies
- computed mass discrepancy $D(r)=V^2_{obs} / V^2_{bar}$
- extracted the outermost measured radius for each galaxy
- classified galaxies as Quiet or Active
- compared D_{outer} across groups
- visualized the separation

The result is a clean, falsifiable, passed test of an ED prediction.

Outcome
This is the first completed astrophysical test of ED.

9. Halo lag in collisions

In high-speed cluster collisions, collisionless dark matter should pass through the interaction region with minimal disturbance. Temporal tension predicts the opposite: because it is a diffusive field tied to activity, it should shear, stretch, or lag behind the baryonic distribution. Lensing peaks should be offset, distorted, or smeared.

10. Correlation with dynamical activity

Dark matter halos depend only on mass. Temporal tension depends on activity. Galaxies with similar baryonic mass but different levels of rotation, turbulence, star formation, or feedback should exhibit different halo strengths.

11. Reduced small-scale substructure

CDM predicts abundant subhalos. Temporal tension, being diffusive, suppresses small-scale clumping. Dwarf satellite counts, strong lensing flux anomalies, and high-resolution rotation curves should reveal smoother profiles than CDM allows.

12. Hysteresis

Temporal tension relaxes slowly. A galaxy that recently experienced a burst of activity should retain an enhanced halo even after the activity subsides. CDM predicts no such memory.

13. Vortex-like distortions

Rotation and shear stir the temporal field, producing swirl-like distortions in the halo. These distortions should appear in lensing maps and in asymmetries between the approaching and receding sides of rotation curves.

14. Asymmetry reflecting baryonic structure

Temporal halos should reflect bars, warps, and asymmetric star formation. CDM halos should remain approximately spherical.

15. Rotation-dependence

Fast-rotating galaxies should host stronger temporal halos, even at fixed mass.

16. Temporal lensing deviations

Temporal halos should produce shear profiles that are smoother and more extended than CDM predictions.

Condensed Matter Tests

17. Mesoscopic Transport Threshold

Introduction

ED predicts that the transition between ballistic and diffusive transport in mesoscopic conductors is governed by a structural threshold in event-density, not by scattering amplitudes or microscopic disorder. As the channel length increases, the ED field crosses a density-gradient limit that forces a non-analytic change in transport behavior.

Standard transport theory predicts a smooth crossover.
ED predicts a sharp, reproducible kink.

This provides a clean, low-cost condensed-matter test of the ED ontology.

Prediction

Conductance
$G(L)$ exhibits a structural transition at a critical length L_{crit} where the event-density gradient saturates:

- below L_{crit}: ballistic-like scaling
- above L_{crit}: diffusive-like scaling
- at L_{crit}: **non-analytic kink in** $G(L)$

This kink is not predicted by standard scattering models, which require smooth interpolation.

The location of L_{crit} depends on the ED-complexity of the channel geometry, not on impurity concentration.

Experimental Path
Fabricate quasi-1D channels with tunable length:

- graphene nanoribbons
- carbon nanotubes
- GaAs/AlGaAs heterostructures
- metallic nanowires

For each device:

1. Measure conductance $G(L)$ across a sweep of channel lengths.
2. Compute ED-complexity from structural geometry.
3. Identify the predicted non-analytic kink at L_{crit}.
4. Compare to standard transport models.

A clean kink at a reproducible ED-complexity threshold is a decisive ED signature.

A smooth crossover with no structural transition falsifies this ED prediction.

18. Superconducting Phase-Stiffness Saturation

Introduction
ED predicts that superconductivity is not a global wavefunction-coherence phenomenon but a high-density event-locking regime. As temperature approaches the critical point, the ED field reaches a saturation limit in its ability to maintain phase-locked participation. This produces a measurable deviation in the temperature-dependence of phase stiffness that standard BCS and KT models do not predict.

Standard theory expects a smooth, monotonic softening of phase stiffness.
ED predicts an early saturation plateau driven by event-density constraints.

This provides a clean condensed-matter test of ED's locality ontology.

Prediction
The superfluid phase stiffness $J(T)$ exhibits a characteristic flattening before the superconducting transition:

- at low temperature: standard stiffness behavior
- approaching T_c: premature saturation of $J(T)$
- near T_c: rapid collapse once the ED-locking regime fails

The saturation temperature T_{sat} depends on ED-complexity (geometry, disorder symmetry, film thickness), not on microscopic pairing strength.

This plateau is not predicted by BCS, KT, or phase-fluctuation models.

Experimental Path

Use thin-film superconductors with tunable geometry:

- NbN
- MoGe
- YBCO
- amorphous InOx

For each sample:

1. Measure phase stiffness $J(T)$ via THz spectroscopy or two-coil mutual inductance.
2. Compute ED-complexity from structural symmetry and thickness.
3. Identify the predicted early-onset saturation plateau.
4. Compare to BCS and KT predictions.

A clear stiffness plateau at an ED-complexity-dependent temperature is a decisive ED signature.

A smooth monotonic softening with no plateau falsifies this ED prediction.

19. Casimir-Gradient Saturation Test

Introduction
ED predicts that near-field forces arise from event-density gradients, not vacuum-mode summation. As two surfaces approach each other in a fluid or dielectric environment, the ED field between them steepens until it reaches a maximum sustainable gradient. Beyond this point, the force no longer follows the standard Casimir–Lifshitz scaling law.

Standard theory predicts a smooth $F \propto 1/d^4$ –type divergence (with material corrections).
ED predicts a saturation regime where the force plateaus as the ED gradient reaches its structural limit.

This provides a clean, high-precision condensed-matter test of ED's density-gradient ontology.

Prediction
The force–distance curve $F(d)$ exhibits three regimes:

- at large separations: standard Casimir–Lifshitz behavior
- at intermediate separations: steepening as the ED gradient builds
- at sub-100 nm separations: force saturation plateau where the ED gradient cannot increase further

The saturation distance d_{sat} depends on the ED-complexity of the cavity (surface roughness, ionic strength, dielectric structure), not on vacuum-mode density.

This plateau is not predicted by any vacuum-mode model.

Experimental Path
Use MEMS/NEMS or microcantilever platforms with tunable plate separation:

- gold–gold surfaces in salt water
- gold–silica surfaces in controlled dielectric fluids
- graphene–substrate cavities
- microfabricated Casimir self-assembly platforms

For each device:

1. Measure force $F(d)$ down to ~10 nm separation.
2. Compute ED-complexity from cavity geometry and dielectric environment.
3. Identify the predicted saturation plateau at d_{sat}.
4. Compare to Casimir–Lifshitz predictions with full material corrections.

A clear force plateau at an ED-complexity-dependent distance is a decisive ED signature.

A smooth divergence with no plateau falsifies this ED prediction.

20. Nonlinear Microresonator Frequency-Phase Threshold

Introduction
ED predicts that nonlinear optical frequency conversion in microresonators is governed not by phase-matching of propagating modes, but by multi-timescale event-density locking inside the cavity. As pump power increases, the ED field inside the resonator crosses a structural threshold that synchronizes frequency and phase across multiple timescales.

Standard nonlinear optics predicts smooth efficiency curves governed by dispersion and mode overlap.
ED predicts a sharp, thresholded jump in conversion efficiency when the ED-locking regime activates.

This provides a clean, high-precision test using existing microresonator platforms.

Prediction
Frequency-conversion efficiency $\eta(P)$ exhibits a structural transition at a critical pump power P_{crit}:

- below P_{crit}: weak, noisy, dispersion-limited conversion
- at P_{crit}: nonlinear jump in efficiency as ED locking activates
- above P_{crit}: stable, high-yield conversion insensitive to small detuning

The location of P_{crit} depends on ED-complexity (cavity geometry, mode multiplicity, material symmetry), not on standard dispersion parameters.

This threshold behavior is not predicted by conventional $\chi^{(2)}$ or $\chi^{(3)}$ models.

Experimental Path
Use high-Q microresonators with tunable pump power:

- chalcogenide micro-racetrack resonators
- SiN microrings

- AlN or $LiNbO_3$ integrated resonators
- UMD-style multi-timescale nonlinear platforms

For each device:

1. Sweep pump power and measure conversion efficiency $\eta(P)$.
2. Compute ED-complexity from cavity geometry and mode structure.
3. Identify the predicted threshold jump at P_{crit}.
4. Compare to standard nonlinear-optical models.

A sharp, reproducible efficiency jump at an ED-complexity-dependent power is a decisive ED signature.

A smooth, dispersion-limited curve with no threshold falsifies this ED prediction.

21. Microresonator Linewidth Asymmetry

Introduction
ED predicts that optical linewidths in high-Q microresonators are shaped not only by material loss and dispersion, but by asymmetric event-density gradients inside the cavity. When the ED field becomes uneven across the resonator—due to geometry, pump detuning, or multi-mode participation—the effective linewidth splits into two distinct branches.

Standard cavity theory predicts symmetric Lorentzian broadening.
ED predicts a structural linewidth asymmetry tied directly to ED-gradient imbalance.

This provides a clean, high-precision photonics test of ED's gradient ontology.

Prediction
The measured cavity linewidth $\kappa(\Delta)$ exhibits:

- symmetric Lorentzian behavior at low ED-gradient
- increasing skew as ED gradients build
- a distinct asymmetric splitting when the ED field crosses a structural threshold

The onset of asymmetry depends on ED-complexity (mode density, cavity geometry, pump configuration), not on standard loss mechanisms.

This asymmetry is not predicted by linear or nonlinear cavity models.

Experimental Path
Use high-Q microresonators with tunable detuning and pump power:

- SiN microrings
- AlN or $LiNbO_3$ integrated resonators
- silica microtoroids
- chalcogenide micro-racetrack resonators

For each device:

1. Sweep detuning and pump power while measuring the cavity transmission spectrum.
2. Compute ED-complexity from mode structure and cavity geometry.
3. Identify the predicted asymmetric linewidth splitting.
4. Compare to standard Lorentzian and nonlinear cavity models.

A reproducible linewidth asymmetry at an ED-complexity-dependent threshold is a decisive ED signature.

A symmetric Lorentzian with no structural splitting falsifies this ED prediction.

22. Photonic Crystal ED-Gradient Limit

Introduction
ED predicts that nanophotonic cavities do not reach arbitrarily high field confinement. As the spatial mode is squeezed, the event-density gradient inside the cavity approaches a structural limit. Beyond this point, further geometric confinement does not increase the local field intensity.

Standard photonic-crystal theory predicts continuous enhancement as the mode volume shrinks.
ED predicts a saturation of field confinement once the ED-gradient limit is reached.

This provides a clean nanoscale test of ED's density-gradient ontology.

Prediction
The cavity field intensity $I(V_{\mathrm{eff}})$ exhibits:

- standard scaling for large mode volumes
- steep enhancement as the cavity approaches the band-edge
- a saturation plateau when the ED gradient reaches its structural maximum

The saturation volume V_{sat} depends on ED-complexity (lattice symmetry, defect geometry, dielectric contrast), not on band-edge dispersion.

This plateau is not predicted by standard photonic-crystal or Purcell-factor models.

Experimental Path
Use nanophotonic cavities with tunable defect geometry:

- Si or SiN photonic-crystal nanobeams
- L3 or H0 defect cavities
- high-contrast 2D photonic-crystal slabs
- inverse-designed nanocavities

For each device:

1. Vary defect geometry to sweep the effective mode volume.
2. Measure local field intensity via nonlinear emission, SHG, or photoluminescence enhancement.
3. Compute ED-complexity from lattice symmetry and defect structure.
4. Identify the predicted saturation plateau at V_{sat}.

Compare to standard Purcell-factor predictions.

A reproducible confinement plateau at an ED-complexity-dependent mode volume is a decisive ED signature.

A smooth, unbounded enhancement with no plateau falsifies this ED prediction.

23. Multi-Timescale Comb Formation Threshold

Introduction
ED predicts that optical frequency-comb formation in microresonators is governed not by smooth nonlinear dispersion dynamics, but by multi-timescale event-density locking. As pump power and detuning are varied, the ED field inside the cavity crosses a structural threshold that synchronizes fast (round-trip), intermediate (thermal), and slow (carrier or lattice) timescales.

Standard Lugiato–Lefever (LLE) theory predicts a continuous transition from modulation instability to soliton formation.
ED predicts a sharp, thresholded onset of stable comb formation when multi-timescale ED locking activates.

This provides a clean, high-precision test using existing microcomb platforms.

Prediction
The comb formation curve $\eta_{comb}(P,\Delta)$ exhibits:

- weak, noisy sidebands at low ED-gradient
- increasing modulation instability as pump power rises
- a sudden transition to stable, broadband combs when ED locking synchronizes the timescales
- robustness to small detuning changes once the ED-locked regime is active

The critical pump power P_{crit} and detuning Δ_{crit} depend on ED-complexity (mode density, cavity geometry, thermal pathways), not on standard LLE parameters alone.

This threshold behavior is not predicted by smooth LLE bifurcation diagrams.

Experimental Path
Use microcomb-capable resonators with tunable pump power and detuning:

- SiN microrings
- AlN or $LiNbO_3$
- integrated resonators
- silica microtoroids
- chalcogenide micro-racetrack resonators

For each device:

1. Sweep pump power and detuning while monitoring comb spectra.
2. Compute ED-complexity from cavity geometry, thermal pathways, and mode structure.
3. Identify the predicted sharp onset of stable comb formation at (P_{crit}, Δ_{crit}).
4. Compare to LLE predictions and standard modulation-instability thresholds.

A reproducible, ED-complexity-dependent threshold for stable comb formation is a decisive ED signature.

A smooth, continuous transition with no structural threshold falsifies this ED prediction.

24. ED-Limited Soliton Step Height

Introduction
ED predicts that the discrete "soliton steps" observed in pumped microresonators are not purely nonlinear-optical bifurcations, but signatures of event-density gradient limits inside the cavity. As pump power increases, the ED field reaches a structural ceiling that caps the maximum achievable soliton energy. This produces a characteristic saturation in the height of the soliton step.

Standard Lugiato–Lefever (LLE) theory predicts that soliton step height scales smoothly with pump power and detuning.
ED predicts a hard upper bound set by the maximum sustainable ED gradient.

This provides a clean, high-precision test using existing soliton-microcomb platforms.

Prediction
The soliton step height $H(P,\Delta)$ exhibits:

- standard growth at low ED-gradient
- steepening as the system approaches soliton formation
- a saturation plateau where further pump power does not increase step height
- robustness of the plateau across small detuning changes once the ED limit is reached

The saturation height H_{sat} depends on ED-complexity (mode density, cavity geometry, thermal pathways), not on nonlinear dispersion parameters alone.

This upper bound is not predicted by standard LLE soliton theory.

Experimental Path
Use microcomb-capable resonators with tunable pump power and detuning:

- SiN microrings
- AlN or $LiNbO_3$ integrated resonators
- silica microtoroids
- chalcogenide micro-racetrack resonators

For each device:

1. Sweep pump power and detuning while monitoring the soliton step in the transmission curve.
2. Compute ED-complexity from cavity geometry, thermal pathways, and mode structure.
3. Identify the predicted saturation plateau in soliton step height.
4. Compare to LLE predictions and nonlinear cavity models.

A reproducible, ED-complexity-dependent saturation height is a decisive ED signature.

A smooth, unbounded increase in step height falsifies this ED prediction.